

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
#include <stdio.h>
#include <stdlib.h>

int main() {

    printf("==== OPERATING SYSTEM INFORMATION ====\\n");
    system("uname -a");

    printf("\\n==== CPU INFORMATION ====\\n");
    system("lscpu");

    printf("\\n==== STORAGE INFORMATION ====\\n");
    system("lsblk");

    printf("\\n==== RUNNING PROCESSES ====\\n");
    system("ps");

    printf("\\n==== SYSTEM RESOURCE USAGE ====\\n");
    system("top -b -n 1");

    return 0;
}
```

Output:

```
pranav@Pranavs-MacBook-Air dir1 % ./partb
==== OPERATING SYSTEM INFORMATION ====
Darwin Pranavs-MacBook-Air.local 25.5.0 Darwin
Kernel Version 25.5.0: Tue Jun  9 22:27:52 PDT 2026;
root:xnu-12377.121.10~1/RELEASE_ARM64_T8112
arm64
==== CPU INFORMATION ====
sh: lscpu: command not found
```

===== STORAGE INFORMATION =====

sh: lsblk: command not found

===== RUNNING PROCESSES =====

PID	TTY	TIME	CMD
36582	ttys000	0:00.10	-zsh
36930	ttys000	0:00.01	./partb

===== SYSTEM RESOURCE USAGE =====

invalid option or syntax: -b

top usage: top

[-a | -d | -e | -c <mode>]

[-F | -f]

[-h]

[-i <interval>]

[-l <samples>]

[-ncols <columns>]

[-o <key>] [-O <secondaryKey>]

keys: pid (default), command, cpu,
cpu_me, cpu_others, csw,
time, threads, ports, mregion, mem,
rprvt, purg, vsize, vprvt,
kprvt, kshrd, pgrp, ppid, state, uid,
wq, faults, cow, user,
msgsent, msgrecv, sysbsd, sysmach,
pageins, boosts, instrs, cycles
[-R | -r]
[-S]
[-s <delay>]

[-n <nprocs>]
[-stats <key(s)>]
[-pid <processid>]
[-user <username>]
[-U <username>]
[-u]
[-W]